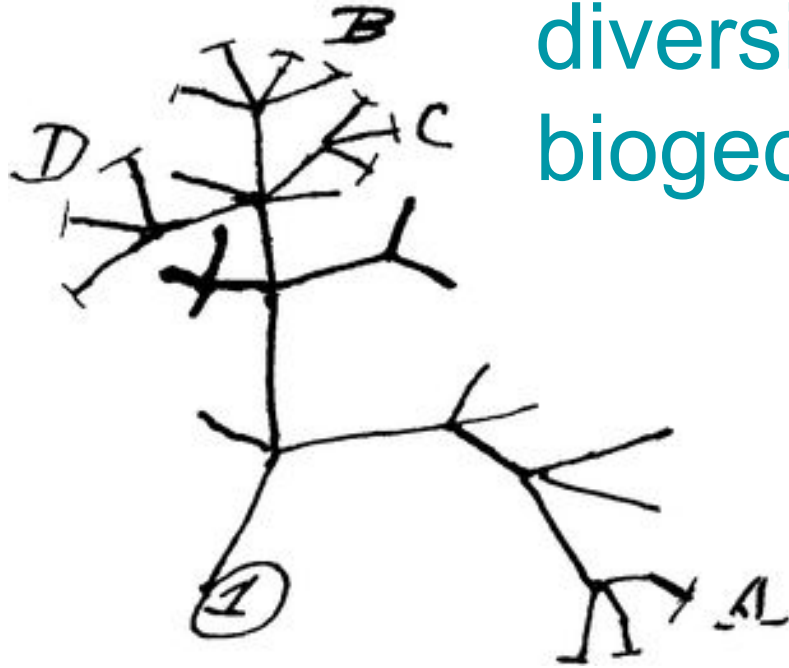


Modelos probabilísticos en  
biología evolutiva:  
diversificación, morfología,  
biogeografía y tendencias

¡BIENVENIDOS!

*I think*



# Temario

## Unidad I - Patrones de diversificación

**1.1 Introducción a la Macroevolución**

**1.2 Introducción a la inferencia bayesiana con RevBayes**

**1.3 Introducción al proceso de diversificación: Estimar la tasa de diversificación de un clado**

**1.4 & 1.5 Estimar tasas de diversificación entre clados y a lo largo del tiempo**

**1.6 Estimar eventos de extinción masiva**

**1.4 Estimar diversificación dependiente de un estado de carácter: modelos de la familia SSE**

# Temario

## Unidad 2: Patrones biogeográficos

**2.1. Modelo DEC uso y limitaciones**

**2.2. Modelo BIB uso y limitaciones**

**2.3. GEOSSE Diversificación dependiente del rango geográfico**

**2.4. Alternativas para estimar eventos de dispersión de larga distancia: ClaSSE**

# Temario

## Unidad 3: Patrones morfológicos

- 3.1 Hipótesis y modelos de evolución morfológica**
- 3.2 Disparidad a lo largo del tiempo**
- 3.3. Reconstrucción ancestral de caracteres**
- 3.4. Factores que afectan las tasas de evolución morfológica**
- 3.5. Evolución puntuada versus gradual**

# Temario

## Unidad 4. Tendencias en Estudios Macroevolutivos

**4.1 Modelos generativos**

**4.2 Aplicaciones de la IA en Estudios Macroevolutivos**

**4.3 Desarrollo de un proyecto de investigación**

# Reglas de convivencia

Queremos que el curso sea un **espacio académico seguro, respetuoso, incluyente e intelectualmente estimulante.**

- Respetemos el trabajo y tiempo de los demás.
  - La clase empezará 10:10 en punto
- Seamos tolerantes de opiniones distintas a las nuestras.
- Seamos curiosos.
  - Participemos en las discusiones.
  - Hagamos preguntas para entender.
  - No hay pregunta tonta, ¡todos estamos aprendiendo!
- ¿algo más?

# Evaluación

|                             |     |
|-----------------------------|-----|
| Exámen                      | 40% |
| Prácticas                   | 20% |
| Participación               | 10% |
| Proyecto Final (Exposición) | 30% |



## El equipo de instructores: Unidad de Síntesis en Sistemática y Evolución (UniSSE)

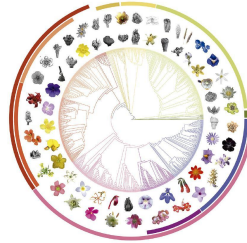


# Cristian Cervantes

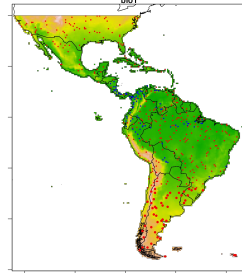
Técnico académico de la Unidad de Síntesis en Sistemática y Evolución (UNISSE)



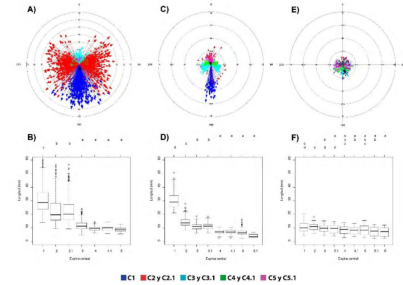
Procesamiento  
y análisis de datos  
genómicos



Filogenética  
comparativa



Análisis geográficos  
y espaciales

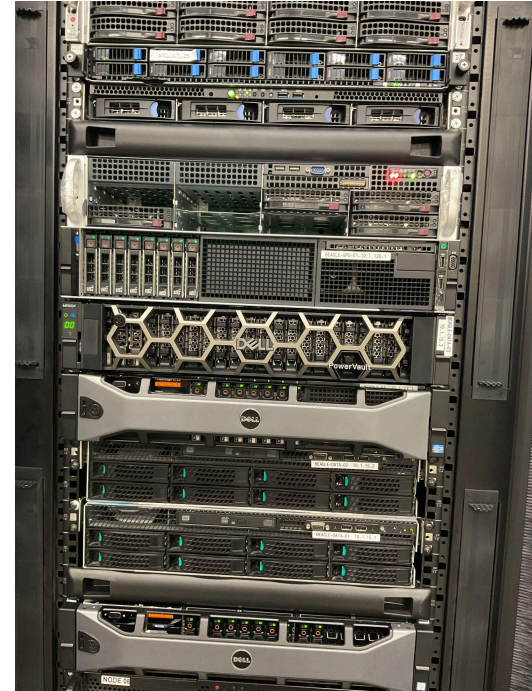


Análisis estadísticos

# Responsable técnico del clúster BEAGLE del Instituto de Biología.



BEAGLE





## El equipo de instructores: Unidad de Síntesis en Sistemática y Evolución (UniSSE)



Karen

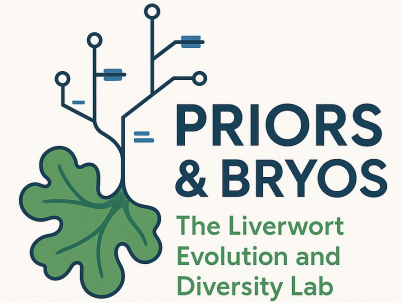
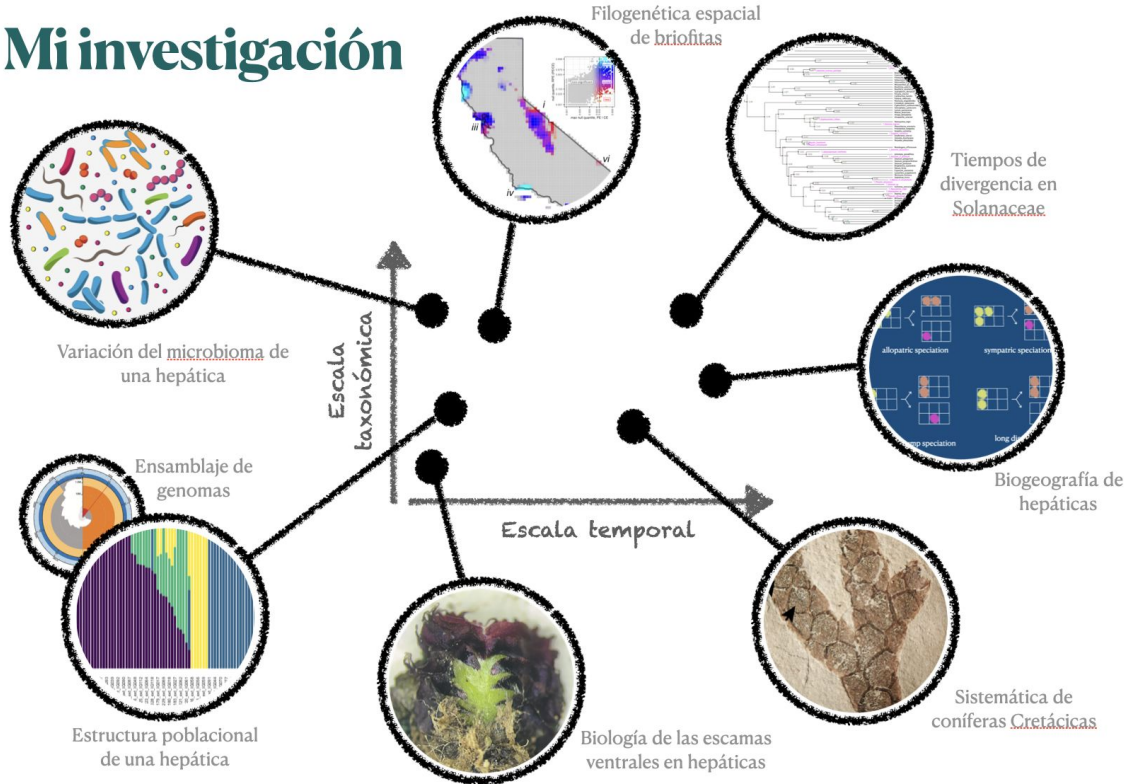


## El equipo de instructores: Unidad de Síntesis en Sistemática y Evolución (UniSSE)



# Ixchel González-Ramírez

## Mi investigación







# Ejemplos de cosas que un modelado cuidadoso te permite hacer/aprender: Solanaceae

*Annals of Botany* 137: 2025–2040, 2026

<https://doi.org/10.1093/aob/mcag011>, available online at [www.academic.oup.com/aob](http://www.academic.oup.com/aob)

ANNALS OF  
BOTANY  
Nonprofit since 1887

## ORIGINAL ARTICLE

### Late Cretaceous origins for major nightshade lineages from total-evidence timetree analysis

Ixchel S. González-Ramírez<sup>1,2,\*†,◉</sup>, Rocío Deanna<sup>3,\*†‡,◉</sup> and Stacey D. Smith<sup>3,◉</sup>

<sup>1</sup>University and Jepson Herbaria and Department of Integrative Biology, University of California, Berkeley, CA 94720, USA,

<sup>2</sup>Missouri Botanical Garden, 4344 Shaw Blvd, St. Louis, MO 63110, USA and <sup>3</sup>Department of Ecology and Evolutionary Biology, University of Colorado, Boulder, CO 80309, USA

<sup>†</sup>Present address: Botany and Mycology Unit, Finnish Museum of Natural History, University of Helsinki, 00100 Helsinki, Finland.

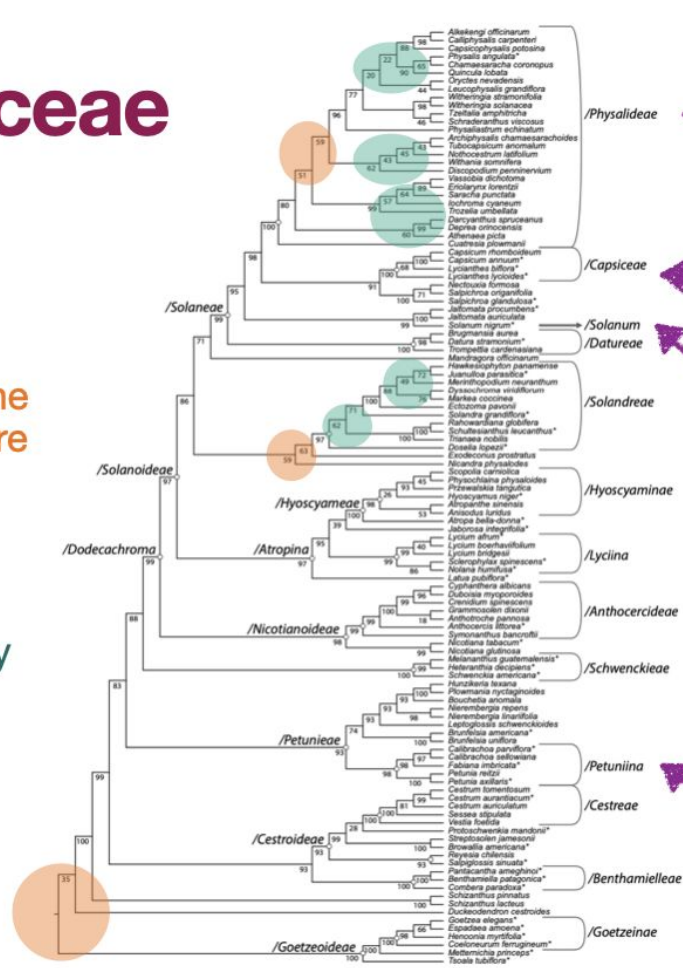
<sup>‡</sup>These authors contributed equally.

\*For correspondence. E-mail [ixchel\\_gonzalezrmz@berkeley.edu](mailto:ixchel_gonzalezrmz@berkeley.edu) or [rocio.deanna@helsinki.fi](mailto:rocio.deanna@helsinki.fi)

Received: 19 July 2025    Returned for revision: 12 November 2025    Accepted: 12 January 2026

# Solanaceae

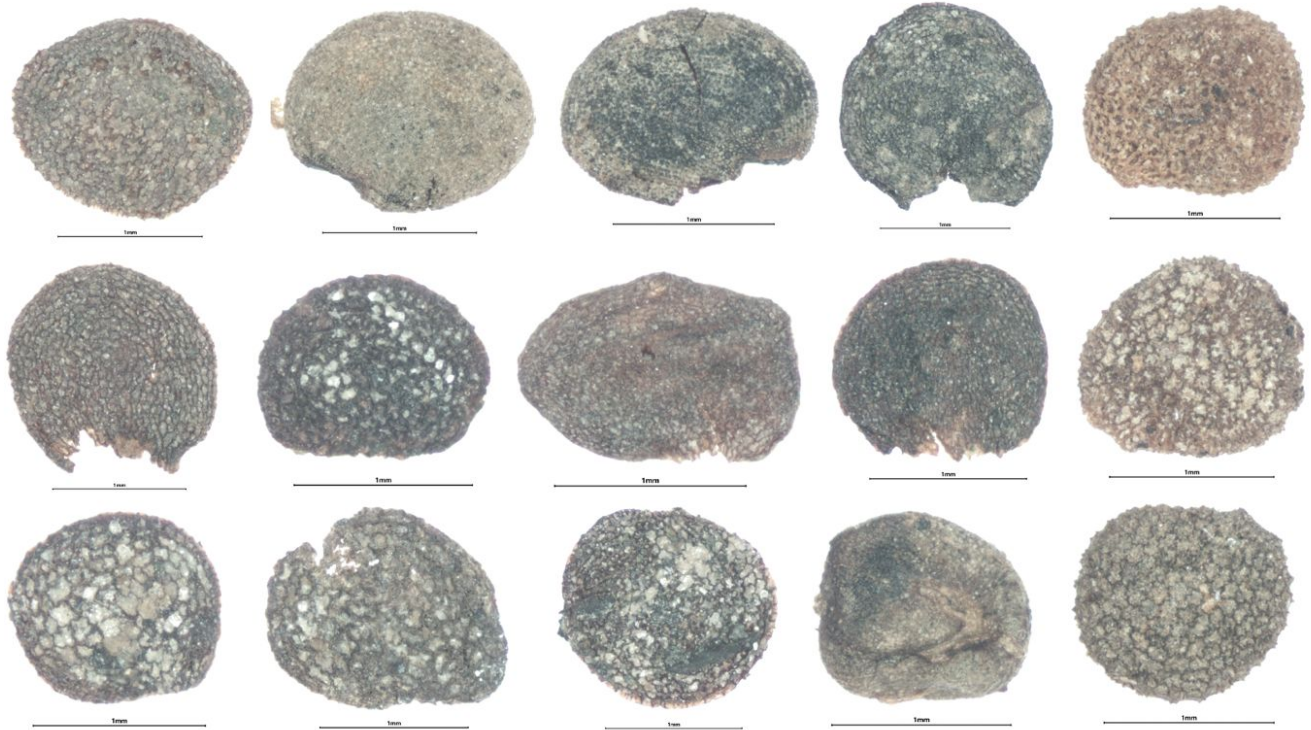
- 2850 spp
- Cosmopolitan
- Some backbone relationships are contentious
- As well as relationships within the berry clade



(Deanna et al., under review)

# Fossil seeds

- 110 fossil seeds
- Geographically widespread



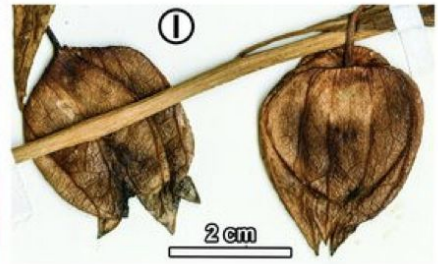
Dorofeev 1951, 1955, 1957, 1966

# Eocene berries

Fruits with inflated five-angled calyces.

**Eocene lantern fruits from Gondwanan Patagonia and the early origins of Solanaceae**

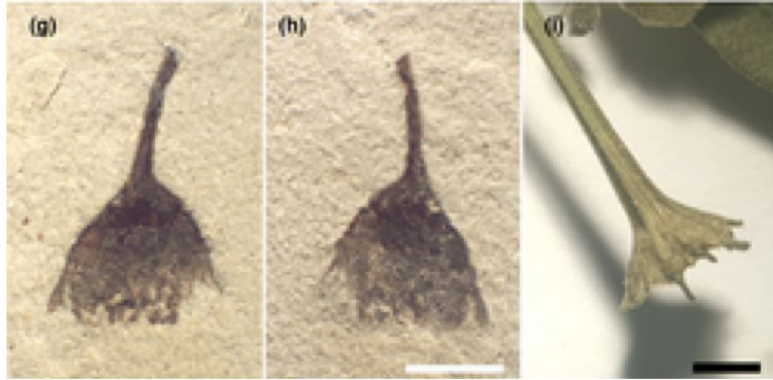
PETER WILF, MÓNICA R. CARVALHO, [...], AND N. RUBÉN CÚNEO +1 authors [Authors Info & Affiliations](#)



*Physalis infinemundi*  
52.2 MYA

(Wilf et al. 2017)

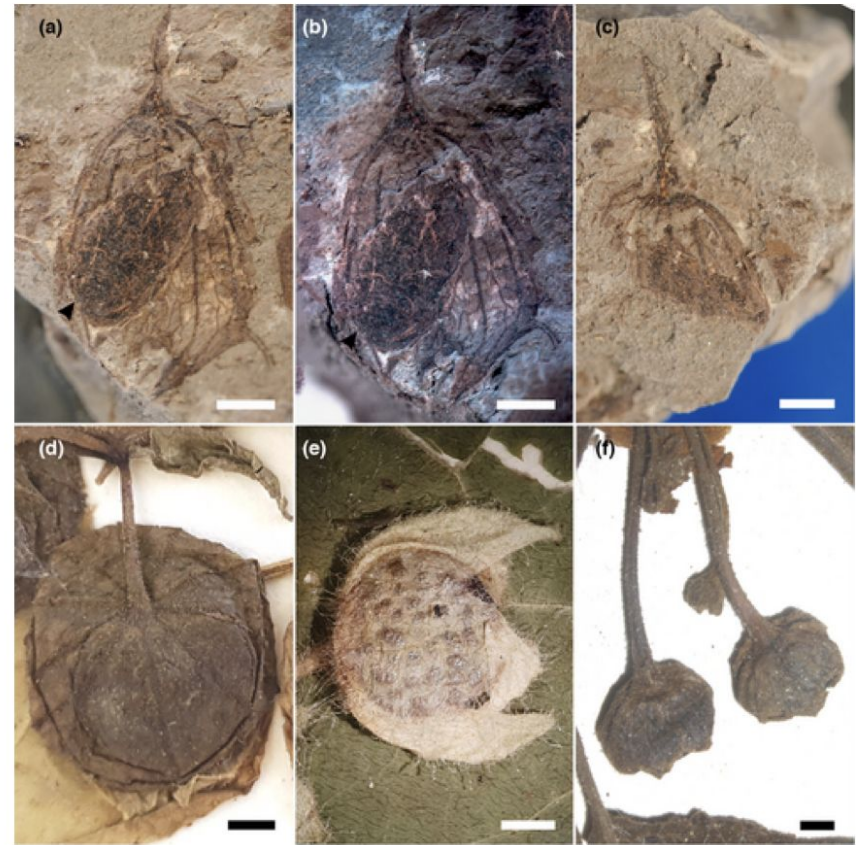
# Eocene berries



*Lycianthoides calycina* (Deanna et al. 2023)  
49.5–51.5Ma

Fossil berries reveal global radiation of the nightshade family  
by the early Cenozoic

Rocío Deanna [✉](#), Camila Martínez, Steven Manchester, Peter Wilf, Abel Campos,  
Sandra Knapp, Franco E. Chiarini, Gloria E. Barboza ... [See all authors](#) >

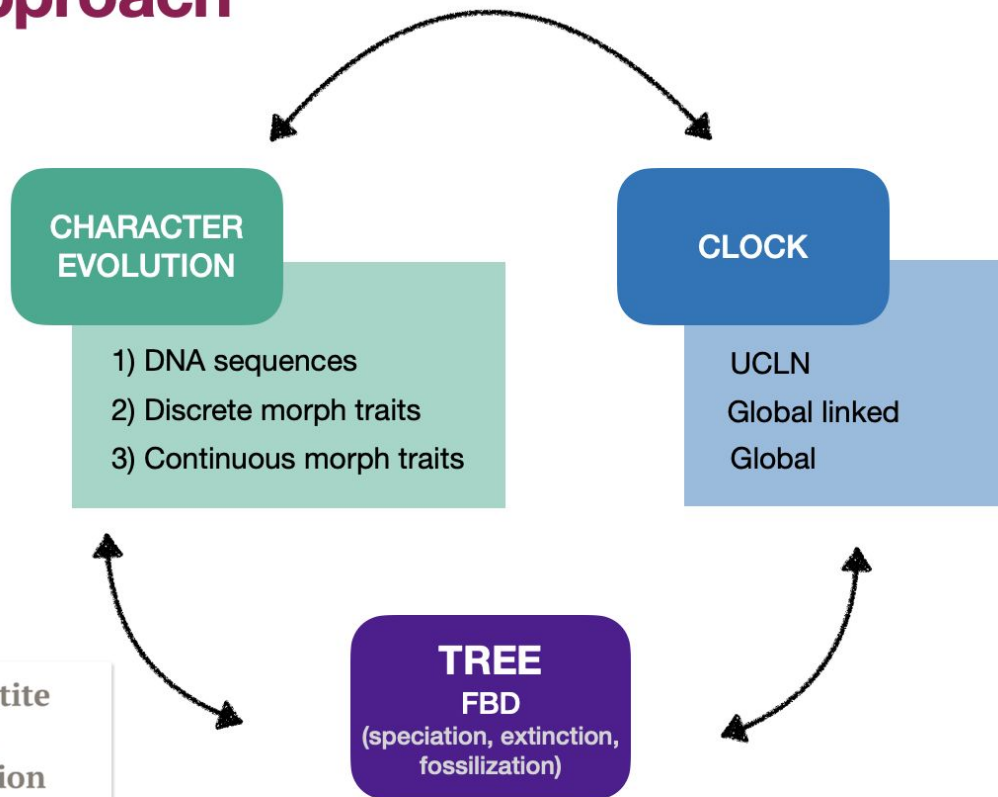


*Eophysaloides inflata*

(Deanna et al. 2023)

# The Bayesian TED approach

- Simultaneously infers topology and divergence times.
- Averages over topological uncertainty.
- Allows the use of multiple fossils.



Understanding the Tripartite  
Approach to Bayesian  
Divergence Time Estimation

Rachel C. M. Warnock  
Friedrich-Alexander Universität Erlangen-Nürnberg  
April M. Wright  
Southeastern Louisiana University

# Data

## 148 taxa

14 fossils

134 extant

## Molecular data

### 10 markers

Nuclear: ITS, LFY, waxy.

Plastid: matK, ndhF, Ndhf-rpl32, psbA, rbcL, trnLF, trnSG

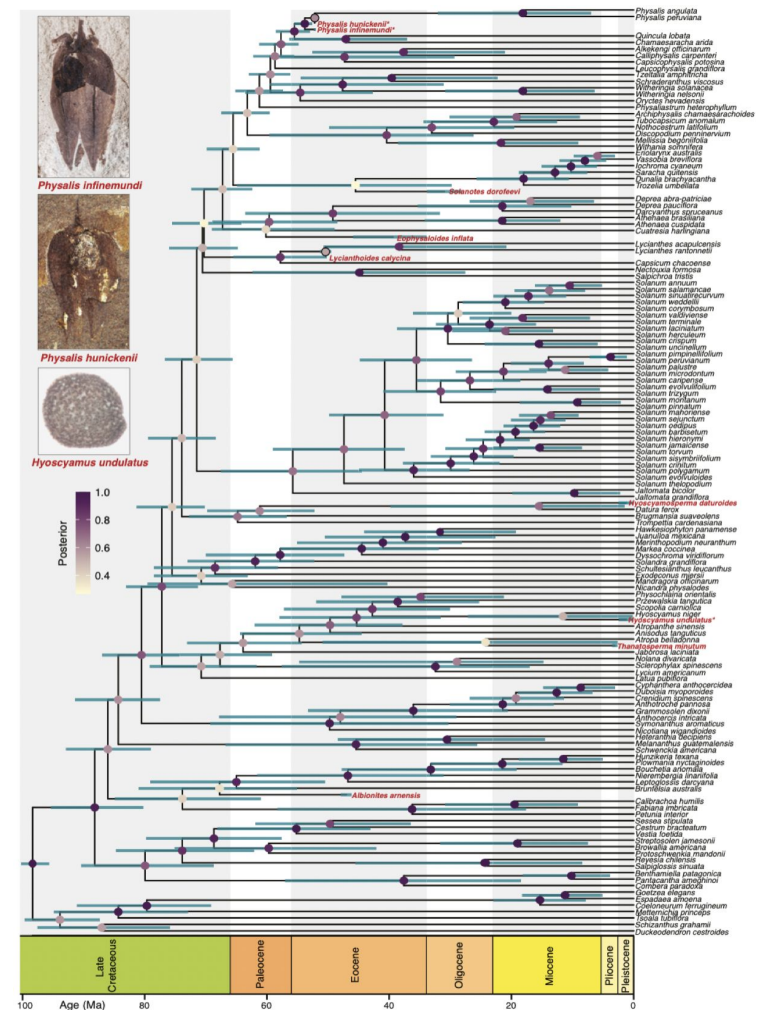
## Morphological data

### 17 categorical and 8 continuous traits

|                         |
|-------------------------|
| 1 compression           |
| 2 hilum_position        |
| 3 embryo                |
| 4 cavity                |
| 5 testal_walls          |
| 6 wings                 |
| 7 elaiosomes            |
| 8 base_invaginated      |
| 9 angled                |
| 10 lobe_sinus           |
| 11 widest_veins         |
| 12 secondary_veins      |
| 13 secondary_veins_fork |
| 14 fruit_type           |
| 15 calyx_inflated       |
| 16 venation_pattern     |
| 17 calyx_teeth          |

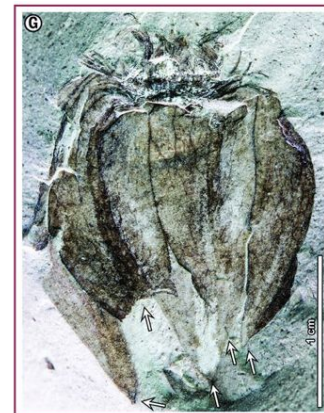
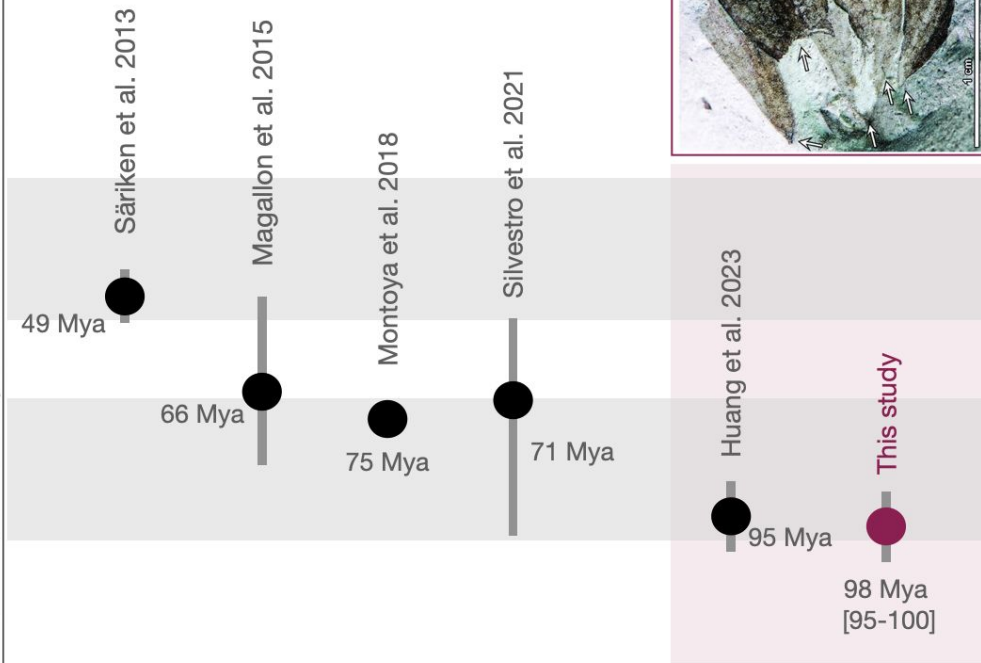
|                         |
|-------------------------|
| 1 seed_length,          |
| 2 seed_ratio,           |
| 3 fruiting_calyx_LT,    |
| 4 fruiting_calyx_ratio, |
| 5 calyx_lobe_LT,        |
| 6 calyx_LT_lobe_ratio,  |
| 7 fruiting_pedicel,     |
| 8 fruit_ratio           |

1) Los tiempos de divergencia, la incertidumbre asociada con esos tiempos y las relaciones filogenéticas.



# How old are Solanaceae?

| Geological Time Scale | Eon        | Era        | System        | Period      | Series Epoch | Stage Age  | Age Ma     |
|-----------------------|------------|------------|---------------|-------------|--------------|------------|------------|
|                       |            |            |               |             |              |            |            |
| Phanerozoic           | Cenozoic   | Quaternary | Holocene      | Upper       | 0.0118       | 0.126      | 0.781      |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Phanerozoic           | Neogene    | Pliocene   | Piacenzian    | Zanclean    | 5.333        | 7.246      | 11.63      |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Phanerozoic           | Miocene    | Langhian   | Serravalian   | Langhian    | 13.82        | 15.97      | 20.44      |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Phanerozoic           | Oligocene  | Chattian   | Rupelian      | Chattian    | 28.1         | 33.9       | 37.8 ±0.5  |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Phanerozoic           | Eocene     | Priabonian | Bartonian     | Lutetian    | 41.2 ±0.5    | 47.8 ±0.2  | 56.0       |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Phanerozoic           | Paleocene  | Thanetian  | Selandian     | Danian      | 59.2         | 61.6       | 66.0 ±0.05 |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Mesozoic              | Cretaceous | Upper      | Maastrichtian | Campanian   | 72.1 ±0.2    | 83.6 ±0.2  | 86.3 ±0.5  |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Mesozoic              | Cretaceous | Lower      | Albian        | Aptian      | 100.5 ±0.4   | 113.0 ±0.4 | 126.3 ±0.4 |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Mesozoic              | Cretaceous | Lower      | Barremian     | Hauterivian | 130.8 ±0.5   | 133.9 ±0.6 | 139.4 ±0.7 |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
| Mesozoic              | Cretaceous | Lower      | Valanginian   | Bernian     | 145.0 ±0.8   |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |
|                       |            |            |               |             |              |            |            |

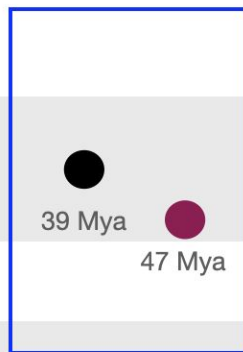


# How old are Solanaceae?

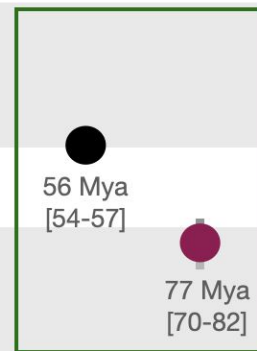
- Implications for internal nodes?

| eonothem<br>Eon | erathem<br>Era | system<br>Period | series<br>Epoch | stage<br>Age  | Age<br>Ma  |
|-----------------|----------------|------------------|-----------------|---------------|------------|
| Phanerozoic     | Cenozoic       | Quaternary       | Holocene        |               | 0.0118     |
|                 |                |                  | Upper           |               | 0.126      |
|                 |                |                  | "Ionian"        |               | 0.781      |
|                 |                |                  | Calabrian       |               | 1.806      |
|                 |                |                  | Gelasian        |               | 2.588      |
|                 |                | Pliocene         | Piacenzian      |               | 3.600      |
|                 |                |                  | Zanclean        |               | 5.333      |
|                 |                | Miocene          | Messinian       |               | 7.246      |
|                 |                |                  | Tortonian       |               | 11.63      |
|                 |                |                  | Serravallian    |               | 13.82      |
|                 |                |                  | Langhian        |               | 15.97      |
|                 |                |                  | Burdigalian     |               | 20.44      |
|                 |                |                  | Aquitanian      |               | 23.03      |
|                 |                | Paleogene        | Chattian        |               | 28.1       |
|                 |                |                  | Rupelian        |               | 33.9       |
|                 |                |                  | Priabonian      |               | 37.8 ±0.5  |
|                 |                |                  | Bartonian       |               | 41.2 ±0.5  |
|                 |                |                  | Lutetian        |               | 47.8 ±0.2  |
|                 |                |                  | Ypresian        |               | 56.0       |
|                 |                |                  | Thanetian       |               | 59.2       |
|                 |                |                  | Selandian       |               | 61.6       |
|                 |                |                  | Danian          |               | 66.0 ±0.05 |
|                 |                | Mesozoic         | Cretaceous      | Maastrichtian | 72.1 ±0.2  |
|                 |                |                  |                 | Campanian     | 83.6 ±0.2  |
|                 |                |                  |                 | Santonian     | 86.3 ±0.5  |
|                 |                |                  |                 | Coniacian     | 89.8±0.3   |
|                 |                |                  |                 | Turonian      | 93.9 ±0.2  |
|                 |                |                  |                 | Cenomanian    | 100.5 ±0.4 |
|                 |                |                  |                 | Albian        | 113.0 ±0.4 |
|                 |                |                  | Lower           | Aptian        | 126.3 ±0.4 |
|                 |                |                  |                 | Barremian     | 130.8 ±0.5 |
|                 |                |                  |                 | Hauterivian   | 133.9±0.6  |
|                 |                |                  |                 | Valanginian   | 139.4 ±0.7 |
|                 |                |                  |                 | Berriasian    | 145.0 ±0.8 |

*Solanum*



*Solanoideae*



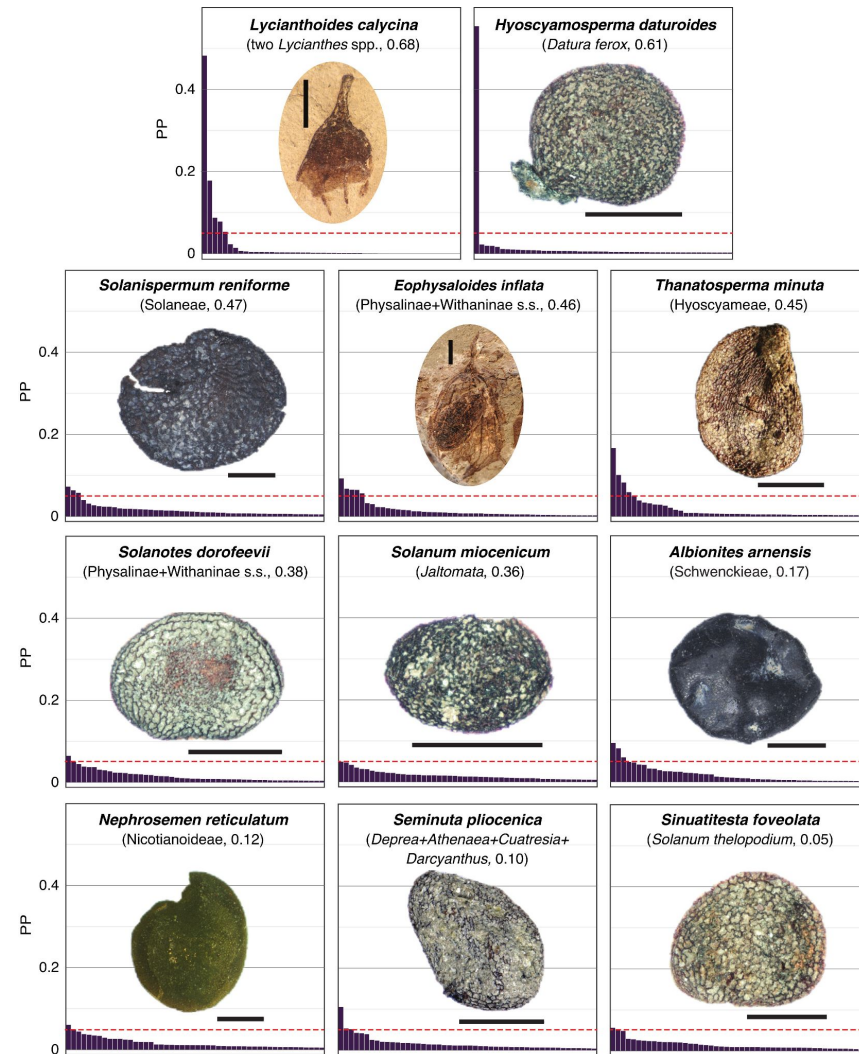
Huang et al. 2023

95 Mya

This study

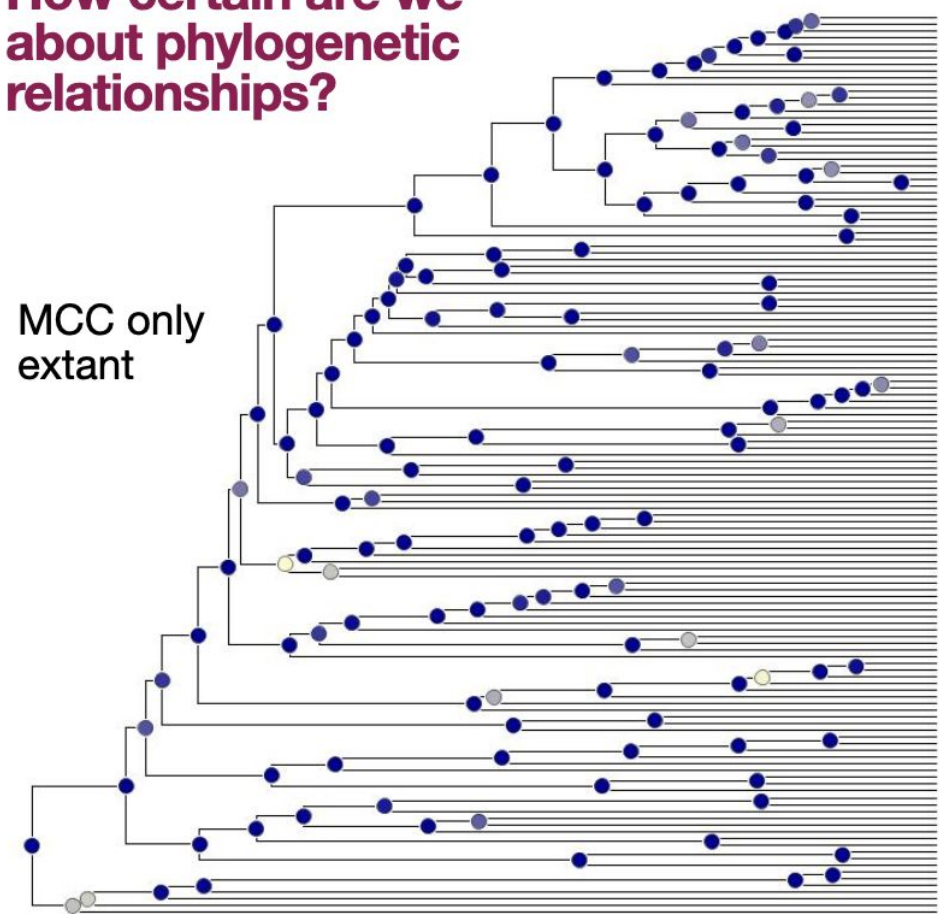
98 Mya  
[95-100]

2) La posición filogenética de fósiles, y la incertidumbre asociada con esa posición.

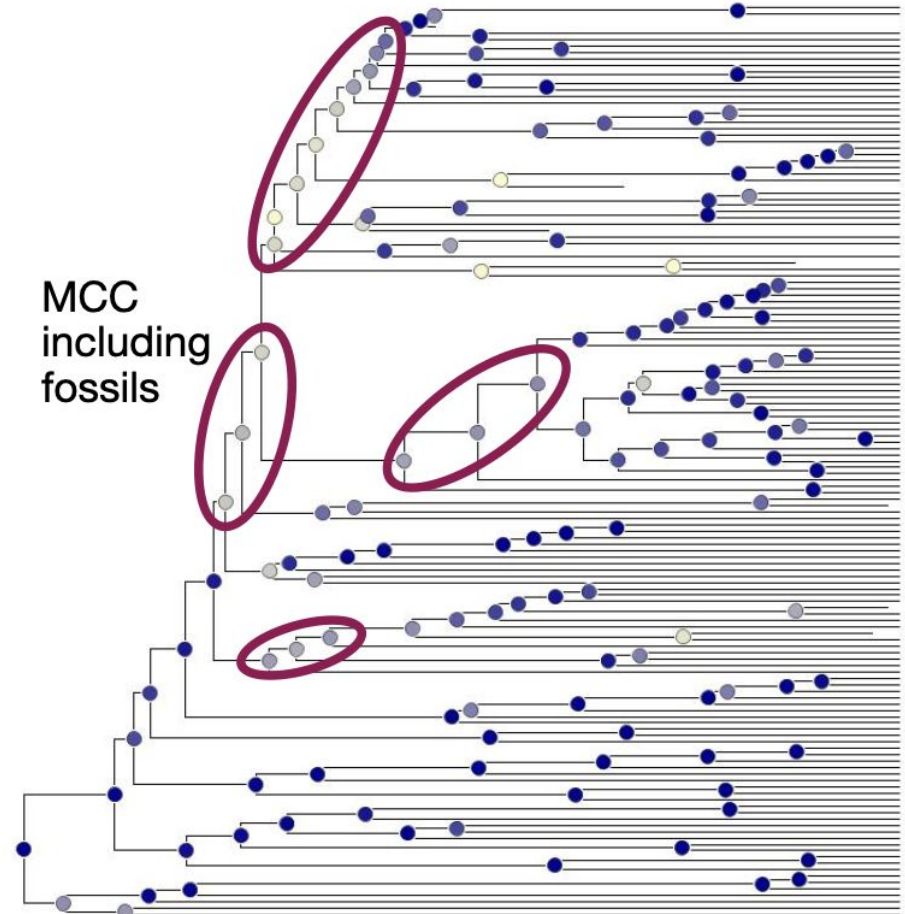


How certain are we about phylogenetic relationships?

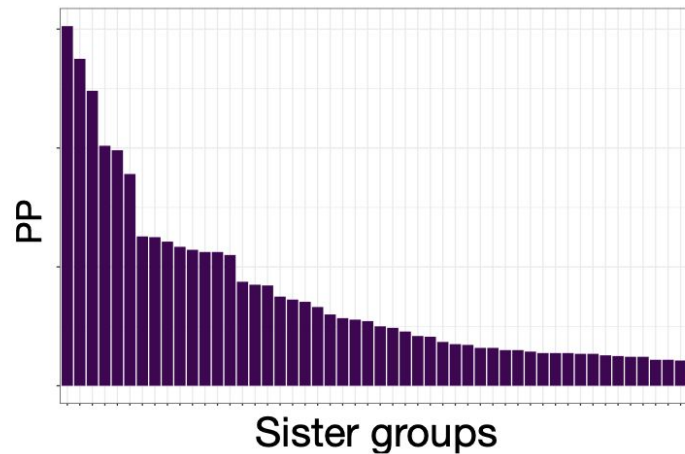
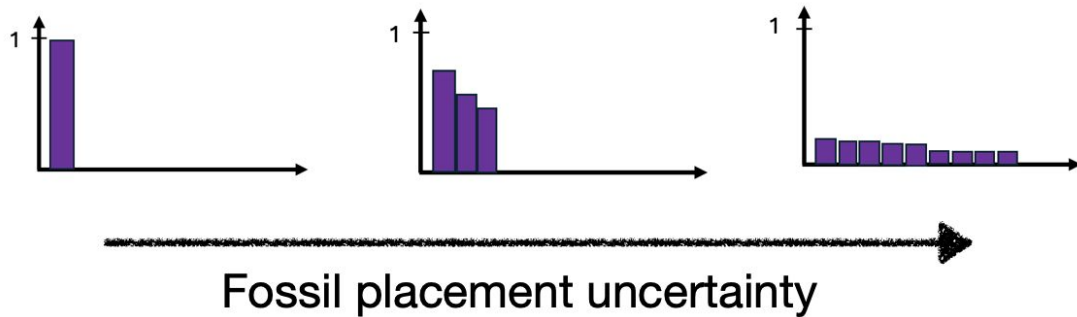
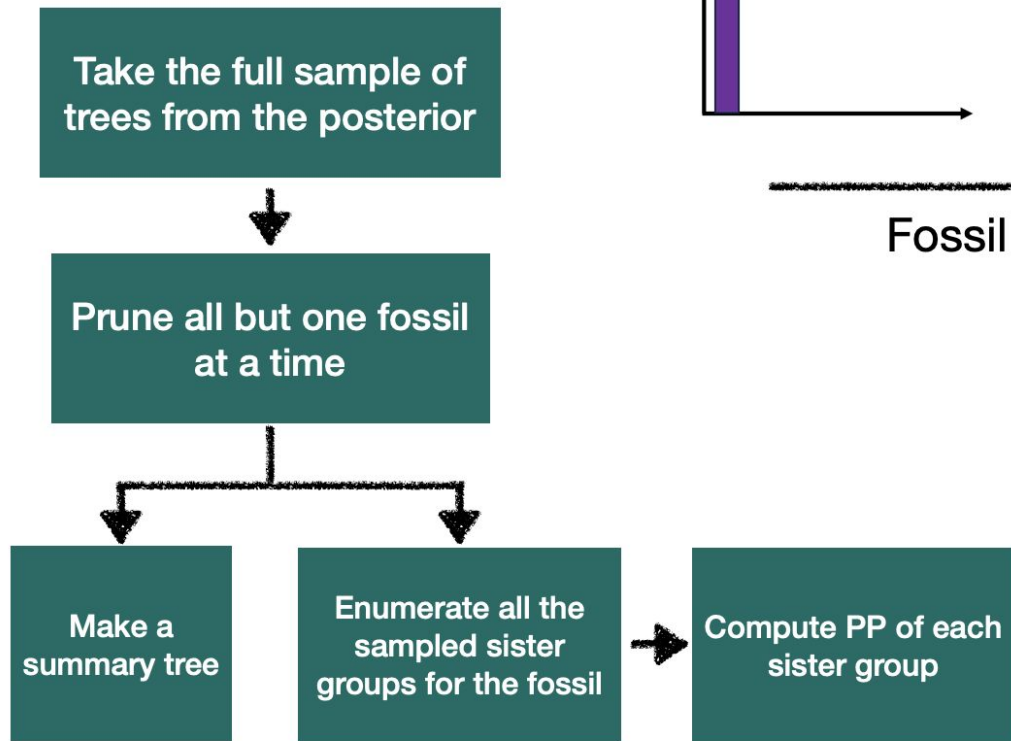
MCC only extant



MCC including fossils



# Fossils



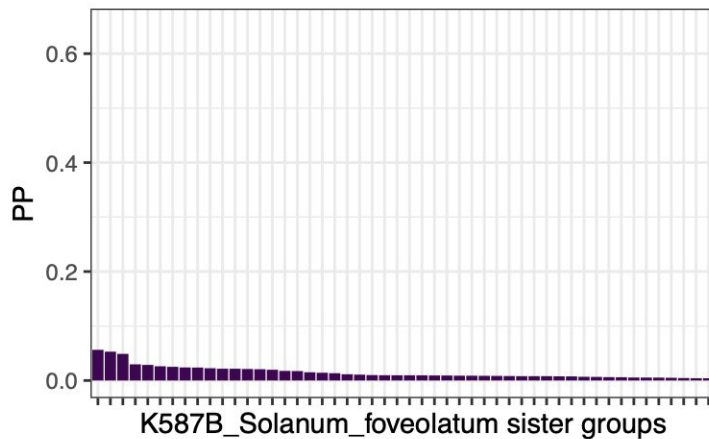
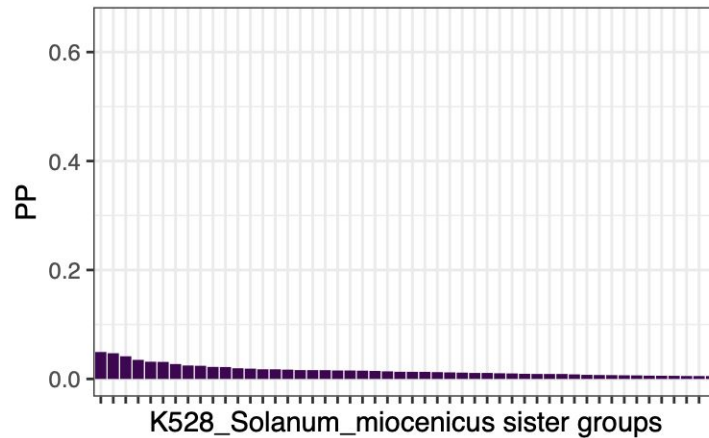
# Fossils



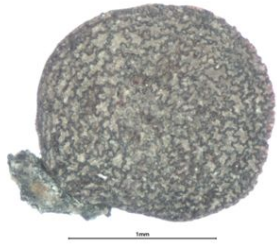
*Solanum miocenicus* (seed) PP = 0.05



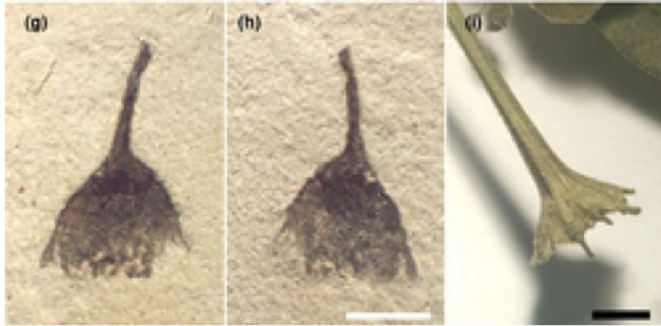
*Solanum foveolatum* (seed) PP = 0.05



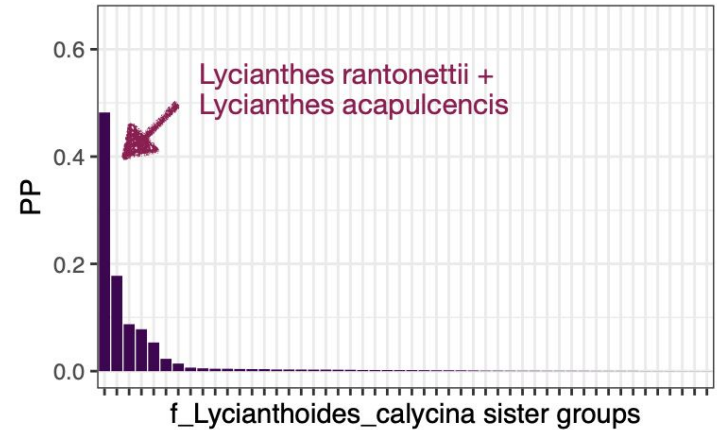
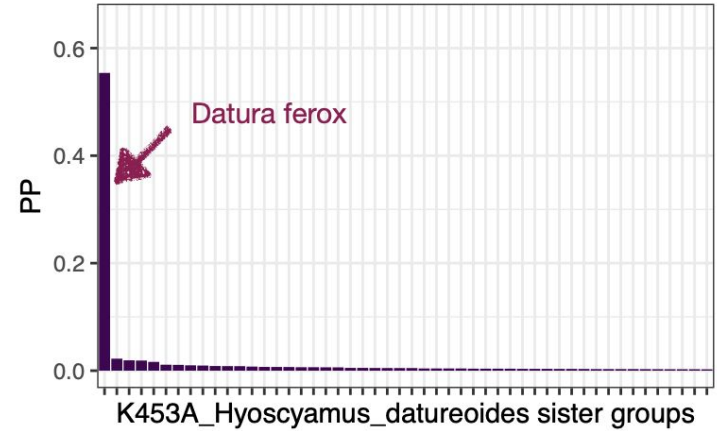
# Fossils



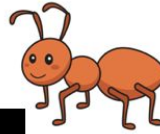
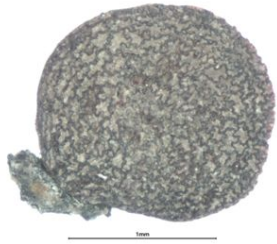
*Hyoscyamus datureoides* (seed) PP = 0.55



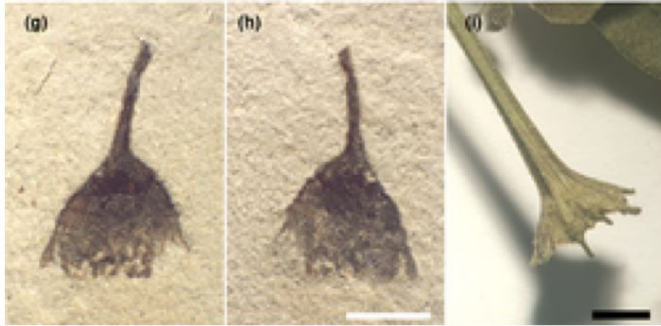
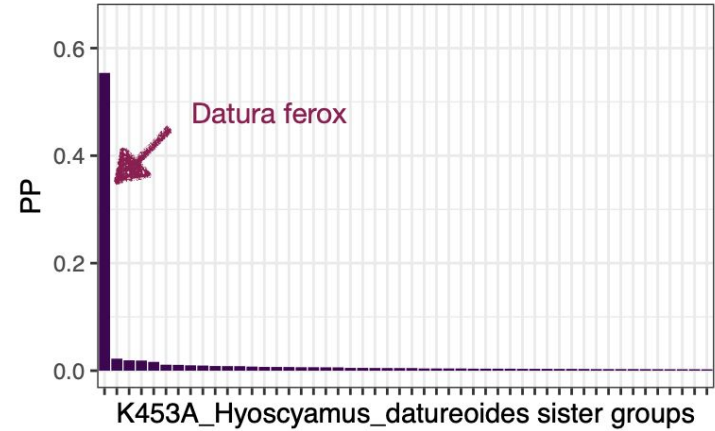
*Lycianthoides calycina* (fruit) PP = 0.48



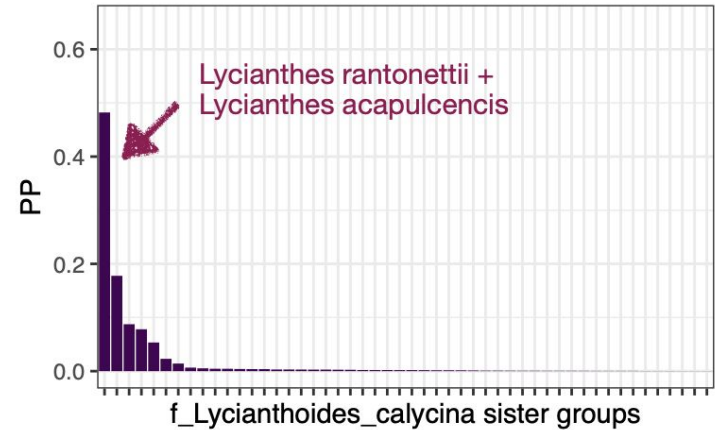
# Fossils



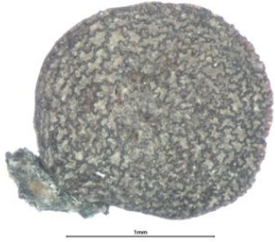
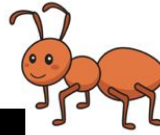
*Hyoscyamus datureoides* (seed) PP = 0.55



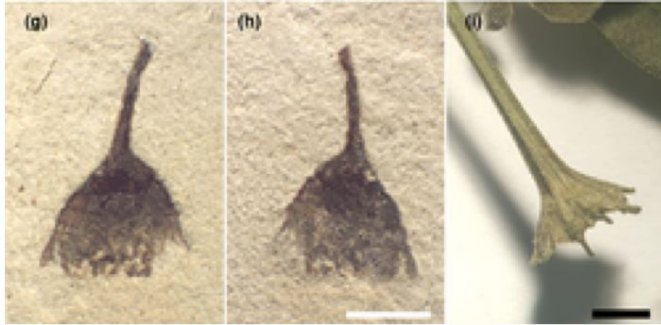
*Lycianthoides calycina* (fruit) PP = 0.48



# Fossils



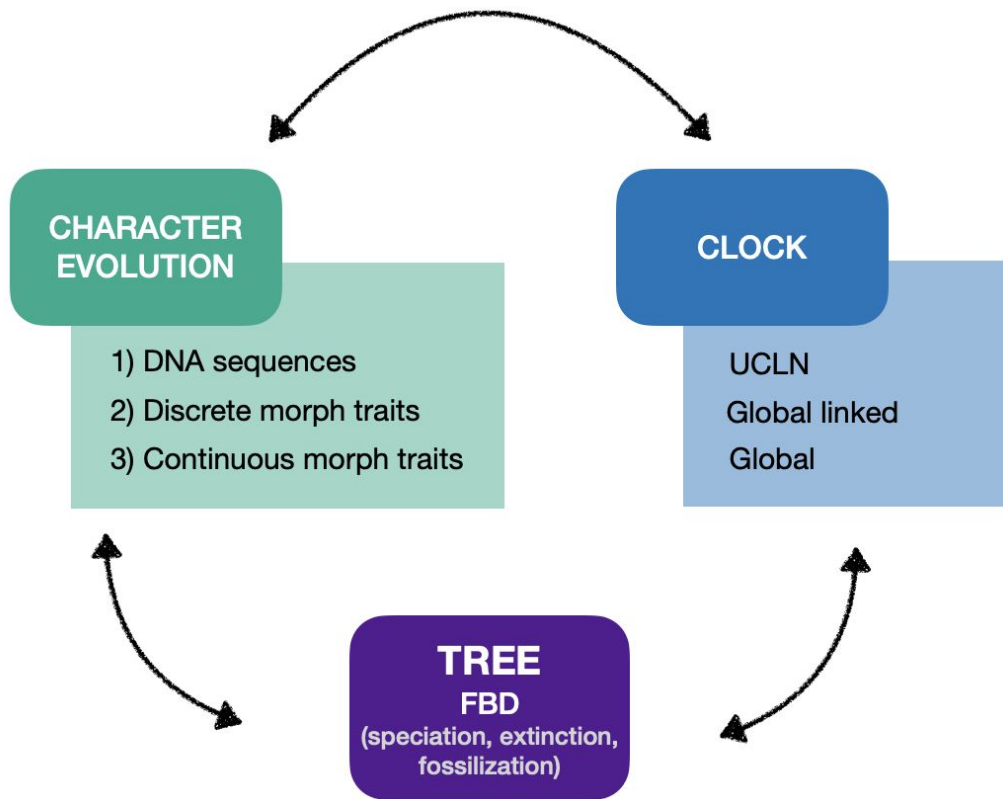
*Hyoscyamus datureoides* (seed) PP = 0.55



*Lycianthoides calycina* (fruit) PP = 0.48

---

3) Retroalimentación  
acerca de patrones  
de diversidad  
morfológica.



Un **modelo generativo** (que modela simultáneamente el proceso que genera el árbol y sus características) nos permite **co-inferir** múltiples parámetros relacionados y nos da un **marco directo** para lidiar con la **incertidumbre**.

# Revisión de software